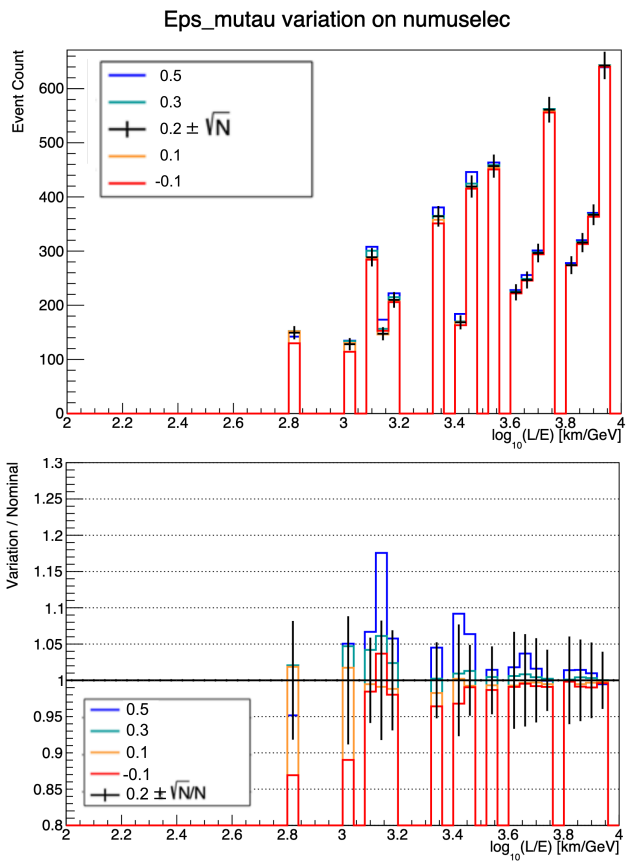
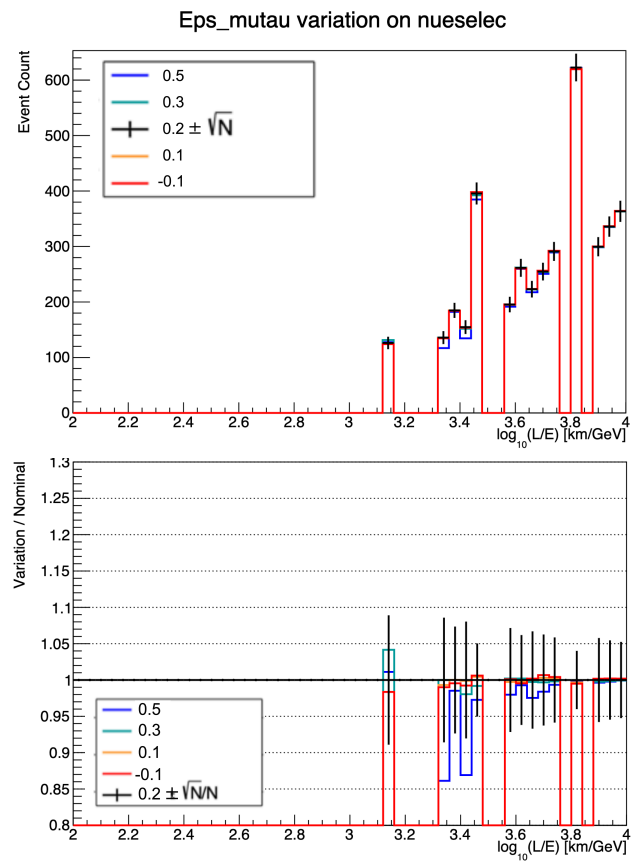
Project 2: DUNE Atmospheric NSI / Bayesian Inference

- Big Physics Question: Do neutrinos experience forces beyond the Standard Model when traversing matter?
- What NSI sensitivity can DUNE's atmospheric program achieve?

NSI Signal Visualization

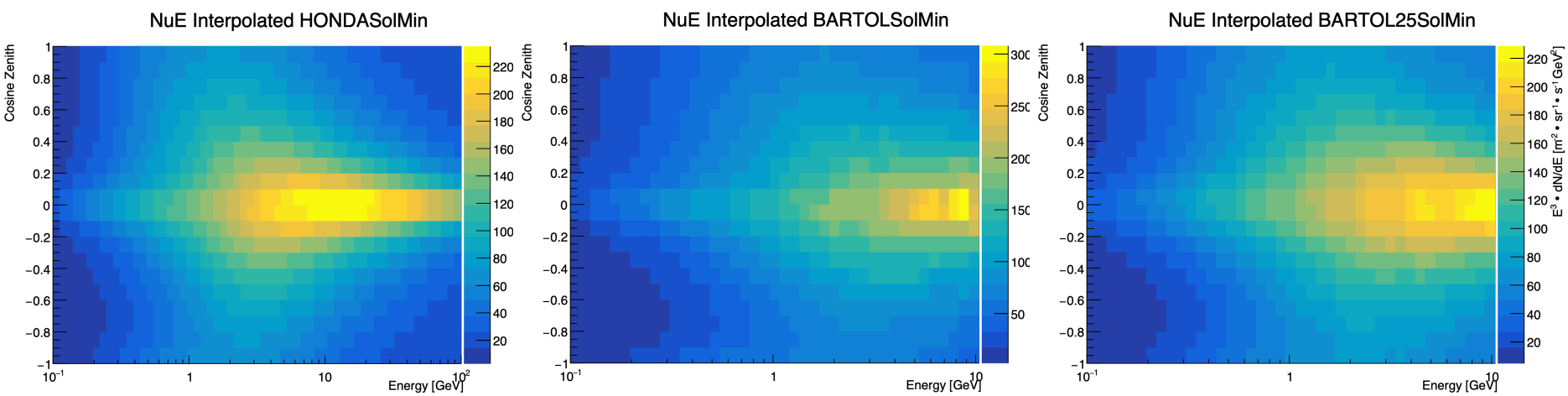
- Example: $\varepsilon_{\mu\tau}$ = -0.1, 0.1, 0.2, 0.3, 0.5
- Visible shift in L/E distribution demonstrate DUNE sensitivity





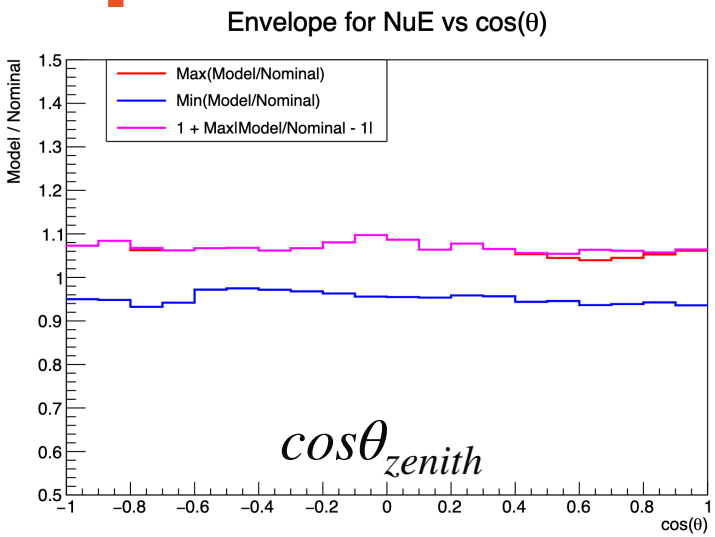
Flux Systematics

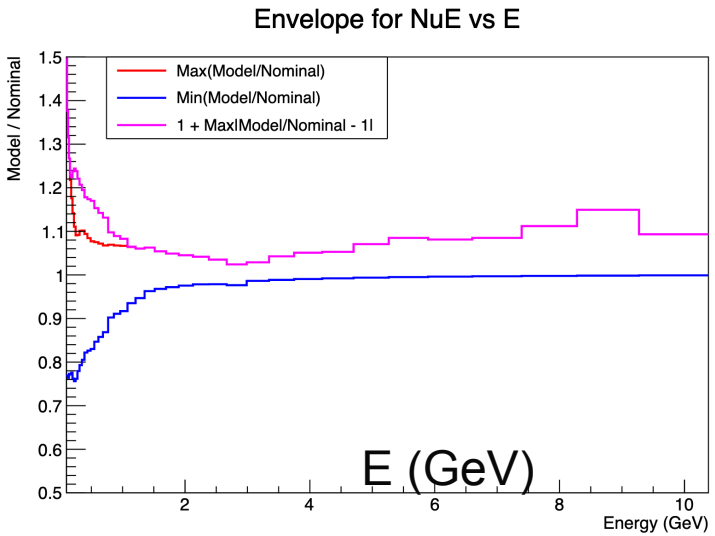
- Goal: Build a realistic NSI sensitivity study for DUNE atmospheric neutrinos
- Challenge: Atmospheric group had no systematics implemented
- Implemented model to model flux uncertainties (Honda vs Bartol)
- Largest uncertainty for atmospheric samples
- Energy & zenith bin normalization



Flux ratio model comparison

- 3 flux models compared:
  - Honda, Bartol, Bartol 2025
- Model differences:
  - Up to 20-30% in energy/zenith regions
- Maximum error envelope approach:
  - Take maximum deviation across all models
  - Conservative systematic treatment

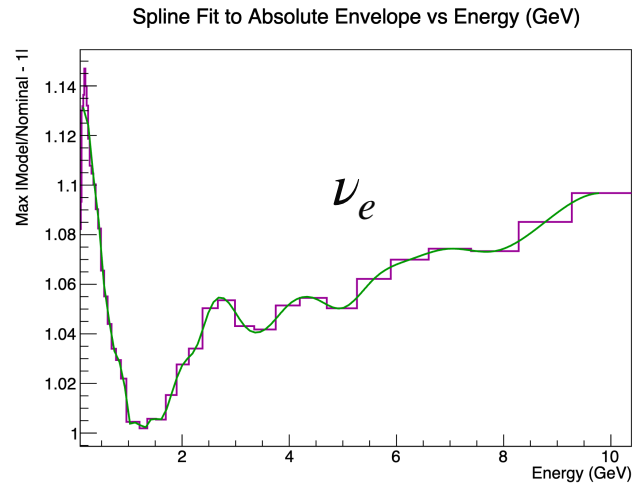


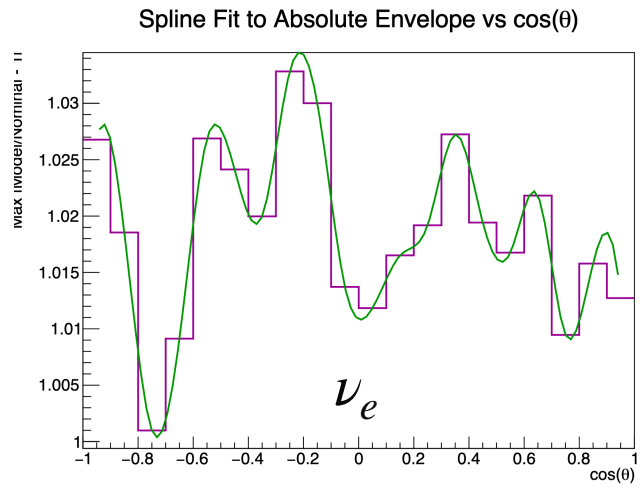


- HONDASolMin as nominal

Spline Parameterization

- Need smooth flux functions for oscillation fits
  - TSpline3 interpolation through envelope
- Implementation in MaCH3:
  - Flux systematics become nuisance parameters
  - Spline nodes constrained by Gaussian prior (mean = 1, σ = 0.1)
  - Fit deforms flux smoothly within allowed uncertainties





Bayesian MCMC Fit Strategy

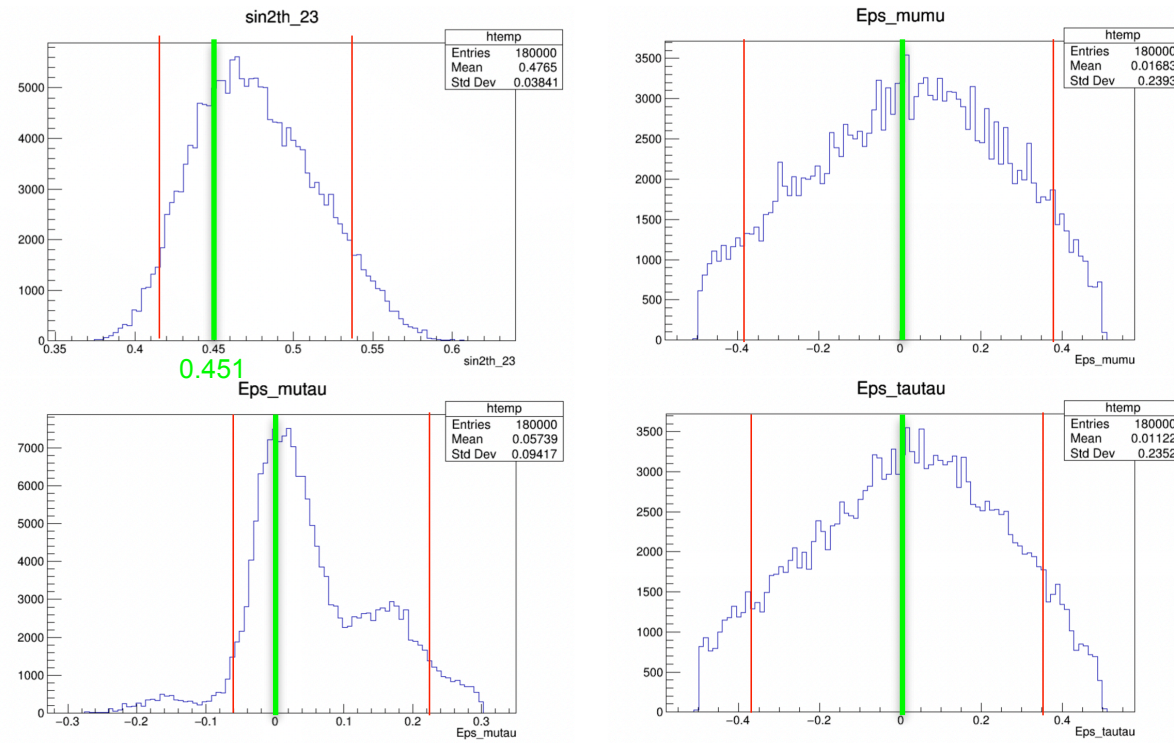
- Markov chain Monte Carlo (MCMC)
- Null hypothesis test: NSI sensitivity (NSI = 0)
  - Generate Asimov data with standard oscillation
  - Question: how far from 0 can we exclude ε?
- Bayesian fit (5 Free parameters)
    - NSI ($\mu - \tau$ sector): $\varepsilon_{\mu\mu}$, $\varepsilon_{\mu\tau}$, $\varepsilon_{\tau\tau}$
    - atmospheric mixing: $sin^2\theta_{23}$
    - Flux systematics: spline-based nuisance parameters
- Fixed parameters:
  - All other oscillation parameters ($\Delta m^2_{ij}, \theta_{12}, \theta_{13}$, etc.)
  - Fixed: e-sector NSI
- Output: 90% credible interval (Bayesian)

DUNE NSI sensitivity - Main Results

- 90% CI using 10 years of simulated atmospheric MC
- $|\varepsilon_{\mu\tau}| < 0.2$ consistent with Super-K constraint
- $sin^2\theta_{23}$ at 0.477 ± 0.038, consistent with global constraints

90% credible limits for NSI parameters (10 years, 400 kt·yr).

Parameter	Constraint
$\lvert\varepsilon_{\mu\tau}\rvert$	< 0.20
$\lvert\varepsilon_{\mu\mu}\rvert$	< 0.39
$\lvert\varepsilon_{\tau\tau}\rvert$	< 0.39



Summary & Conclusion

- NOvA Results:
  - First atmospheric neutrino analysis in NOvA
  - Surface location -> 0.014% ν_μ purity
  - ν_e purity: 3.06%
- DUNE prospects:
  - $|\varepsilon_{\mu\tau}| < 0.2$ (consistent with current limits)
  - Strong $\varepsilon_{\mu\mu}$, $\varepsilon_{\tau\tau}$ degeneracy limits diagonal NSI
- My contribution
  - Developed NOvA atmospheric selection from scratch
  - Built flux systematic framework for DUNE atmospheric
  - First full DUNE atmospheric NSI sensitivity study